

CPSC 483: Introduction to Machine Learning

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Cal State Fullerton

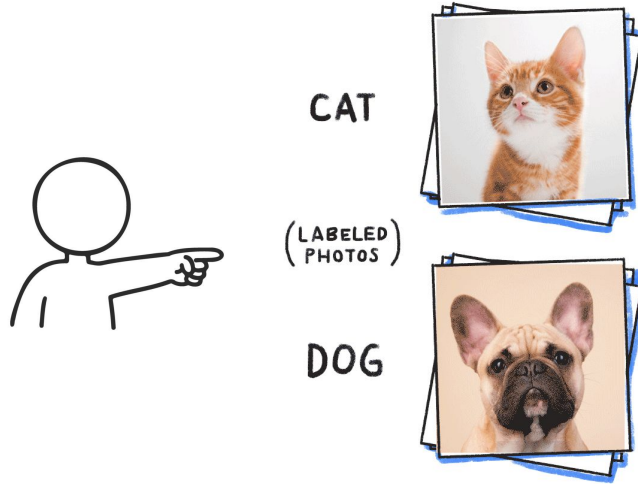


SEMI-SUPERVISED LEARNING

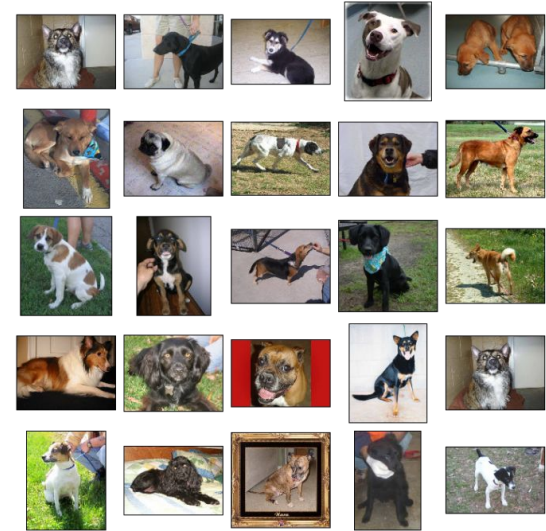
Semi-Supervised Learning - Why?

Lack of training data in real-world

- Labeled data (training data) is hard to acquire
 - Human annotation is time consuming, and labeling requires experts. **Expensive!**



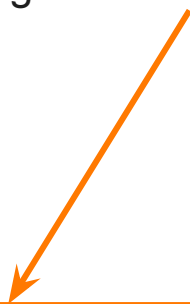
- Unlabeled data is abundant
 - Automatically measured data or databases. **Cheap!**



Semi-Supervised Learning (aka. Weak Supervision)

A branch/paradigm of machine learning that combines supervised and unsupervised algorithms by using both labeled and unlabeled data

- Unlabeled examples used in model training **MUST be relevant** to the task the model is being trained to perform



Formally speaking...

- Semi-Supervised Learning requires that the **distribution $p(x)$** of the input data must contain information about the **posterior distribution $p(y|x)$**
 - that is the conditional probability of a given data point (x) belonging to a certain class (y)

Semi-Supervised Learning - *Inductive Learning Methods*

Aim to **directly** train a classification (or regression) model, using both labeled and unlabeled data

- makes use of a small amount of labeled data to create a training data set with a large amount of unlabeled data

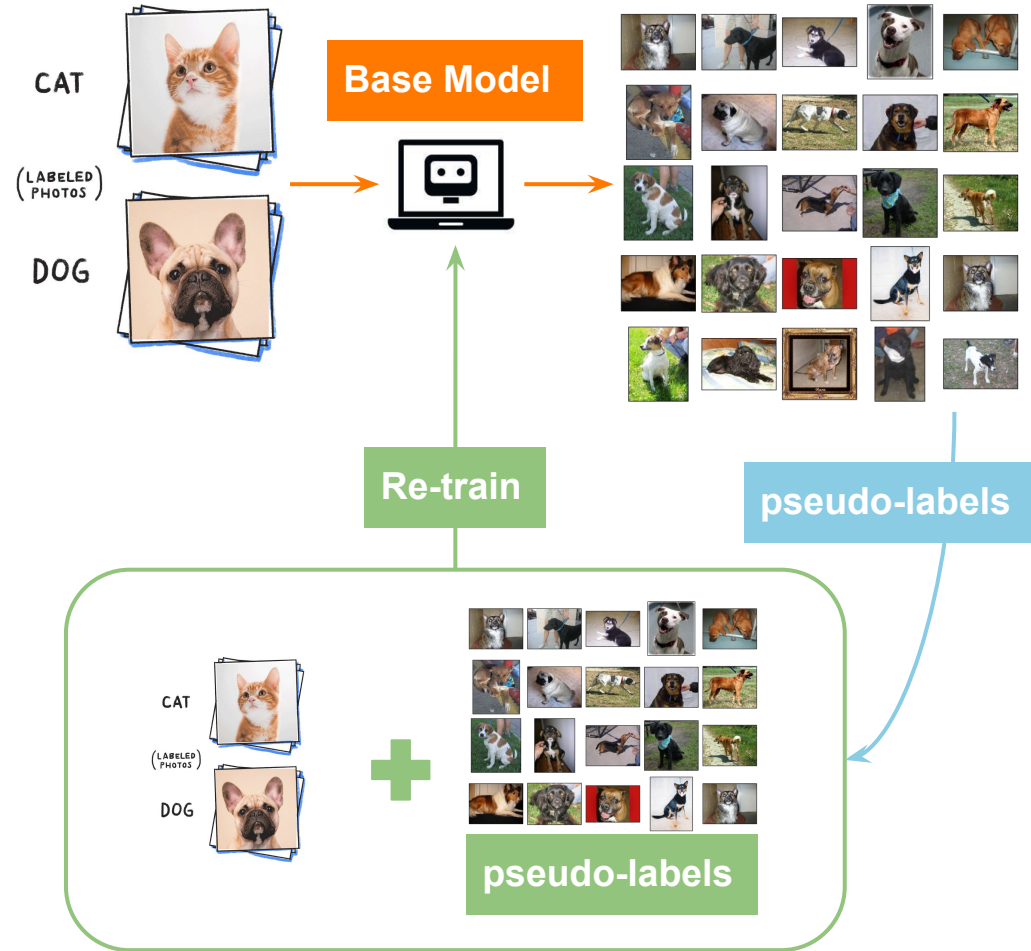
Inductive Semi-Supervised Learning methods can generally be differentiated by the way in which they incorporate unlabeled data via:

- a pseudo-labeling step
- an unsupervised pre-processing step
- or by direct incorporation into the model's objective function

Semi-Supervised Learning

Self-training (via *pseudo-labeling*)

- A supervised learning algorithm is trained based on the labeled data only to have a classifier (*base model*)
- The classifier is applied to the unlabeled data to generate labeled examples (like clustering)
- Feed those new labeled samples and original labeled samples as training data to the supervised learning algorithm to learn a better model
 - Generally, only the labels the classifier is most confident of are added at each step



Semi-Supervised Learning

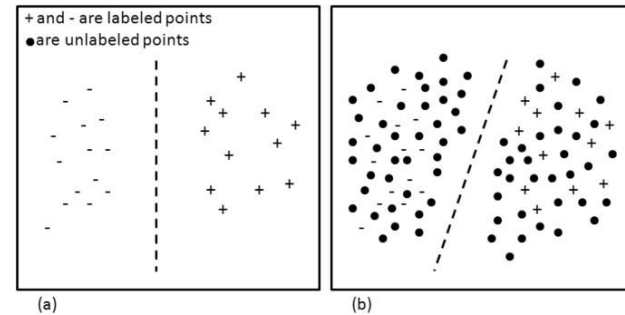
Unsupervised pre-processing

- Like wrapper methods, such techniques can essentially be used for any supervised base learner
- But in contrast to wrapper methods, the “main” supervised model is ultimately trained only on originally (human-annotated) labeled data points
- Unsupervised (or self-supervised) pre-training allows models to learn **useful representations** of the input space, reducing the amount of labeled data needed to fine tune a model with supervised learning
 - Commonly learn through auto-encoder (a type of neural network)

Semi-Supervised Learning - Others

Clustering-Based (i.e., *cluster-than-label*)

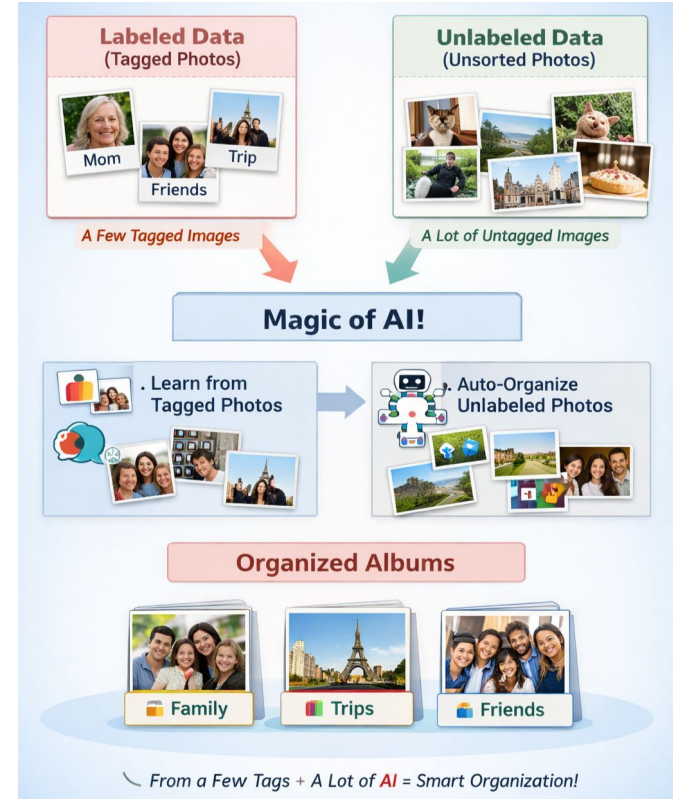
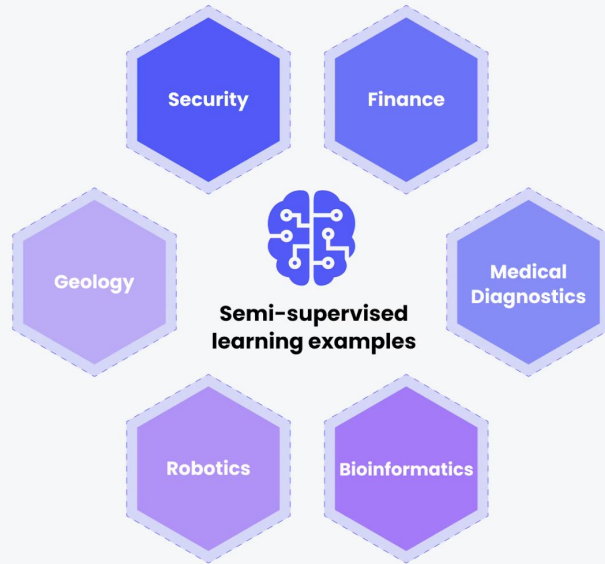
- Group data points based on their similarities
- If a cluster contains mostly labeled data of a certain class, unlabeled data in that cluster can be assigned the same label through inference



Active Learning

- A type of transductive learning
 - Aim only to yield label predictions for unlabeled data
- Used to determine which unlabeled samples would provide the most helpful information if manually labeled

Semi-Supervised Learning - *Applications*



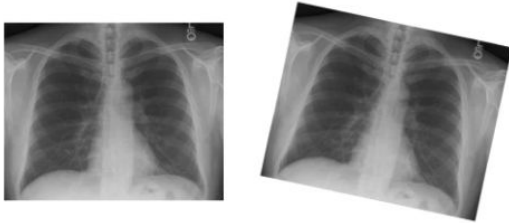


SELF-SUPERVISED LEARNING

Self-Supervised Learning

Idea

- Use the labels that are naturally part of the data or use unsupervised learning ideas to generate supervisory signals rather than relying on labeled data



Single-image instances with augmentation



Multiple views of the same patient (spatial)



Self-Supervised Learning

Pretext Tasks

- Define a task based on the data itself
- To learn meaningful representations of unstructured data
- Those learned representations can be subsequently used as input to a **downstream task**
 - i.e., like a supervised learning task or reinforcement learning task
 - The reuse of a pre-trained model on a new task is referred to as “transfer learning”

Examples

- Running text as training data
 - From a sentence, “The cat sat on the mat.”, randomly mask a word “The cat [] on the mat.”, then train the model to predict the masked word
- Computer vision
 - Image annotation and classification, object detection, semantic segmentation, etc.
- Representation learning
 - Autoencoders (encoding-decoding)

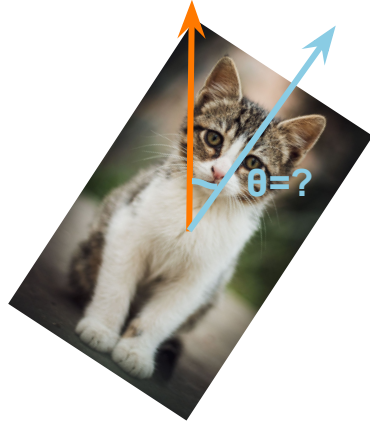
Self-Supervised Learning - *Pretext Tasks*

Capabilities of Self-Supervised Learning

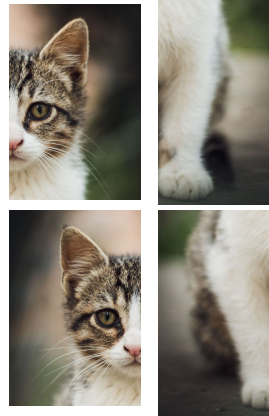
- Predict any part of the input from any other part
- Predict the future from the past or the past from the present
- Predict the obscured from the visible and vice versa
- Pretend there is a part of the input you don't know and predict that



Image Completion



Rotation Prediction



"Jigsaw Puzzle"



Colorization

Self-Supervised Learning vs. Others

Self-Supervised Learning vs. Supervised Learning

- Both optimize predictions against a **known target**
- Supervised learning requires human-labeled data
- Self-supervised learning generates targets from the data itself

Self-Supervised Learning vs. Unsupervised Learning

- Both can learn from **unlabeled data**
- Unsupervised learning discovers patterns or structure
- Self-supervised learning creates a specific prediction task

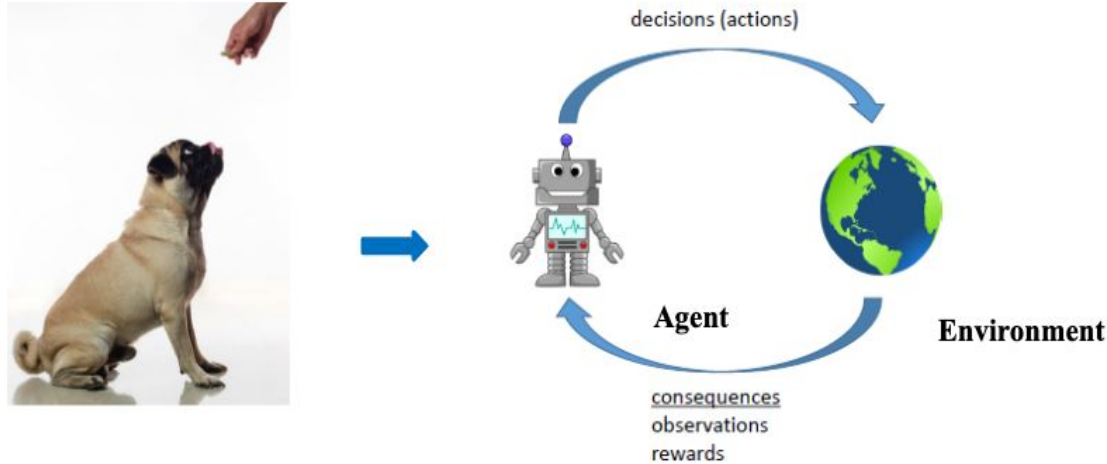
Self-Supervised Learning vs. Semi-Supervised Learning

- Semi-supervised learning leverages a small labeled dataset and a large amount of unlabeled dataset
- Self-supervised learning creates supervision directly from unlabeled data



REINFORCEMENT LEARNING

Reinforcement Learning



RL is a goal-directed learning through **a series of actions** that result in consequences as **rewards**

- The **agent** adapts by exploring a variety of actions while progressively favoring those actions that produce the most short-term reward (**exploiting**), ultimately aiming for the maximum long-term reward

Note that the training data can be used to build an initial model but NOT required

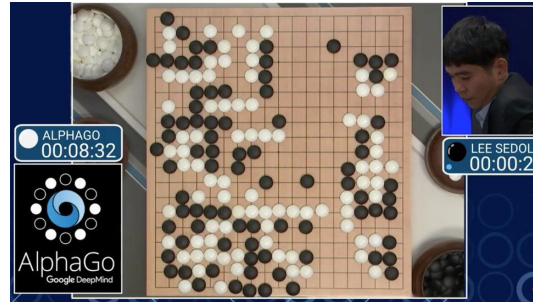
Reinforcement Learning - *Applications*

Robot Walking

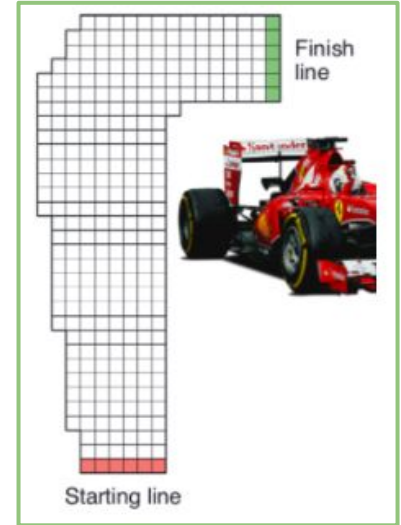


Playing Go games

- AlphaGo beats the human Go champion, Se Dol Lee.
- AlphaZero can even learn the game rules (self-taught) and beat both AlphaGo and the human champions



Autonomous vehicles



Reinforcement Learning Methods

Model-based reinforcement learning

- Dynamic programming to solve the Bellman optimality equations through
 - Policy iteration
 - Value iteration

Model-free reinforcement learning

- Monte-Carlo method
- Temporal-difference
- State-Action-Reward-State-Action (SARSA)
- On-policy and off-policy methods
- Q-Learning
 - Deep Q-learning Network (DQN)

Is RL Supervised or Unsupervised?

- RL is a supervised learning method
 - **Supervised**: A policy can be improved and optimized by reward experience
- RL is an unsupervised learning method
 - **Unsupervised**: Explicit goals are not given but forced to learn those optimal goals by trial and error

Therefore, RL is **neither** supervised **nor** unsupervised but **both**

- The agent keeps on adapting by exploring a variety of actions while progressively exploiting (favoring) those actions producing the most reward
- The best action is taken by exploiting a reward through the improved policy

OTHER CATEGORIES OF MACHINE LEARNING

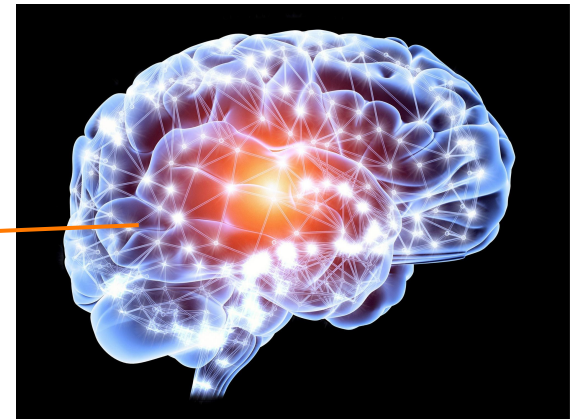
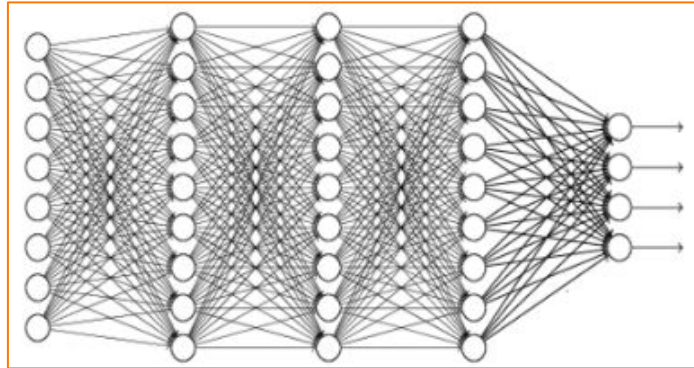
Deep Learning

Deep learning is **NOT a separate learning paradigm** like supervised or reinforcement learning

- Rather, it's a technique/model/architecture (using neural networks) that can be applied within various paradigms

Definition

- Deep Learning is an advanced subset of ML that uses many layers of processing to **simulate the human brain's** decision making process



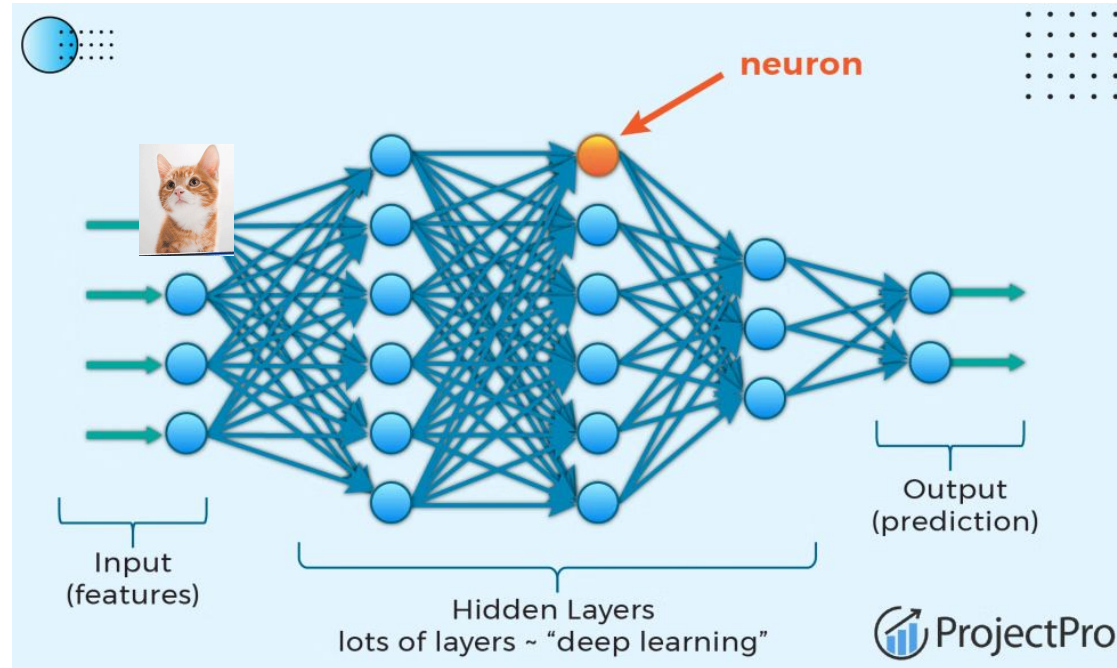
Deep Learning

Uses multiple layers between the input and output

- Input layer
- Multiple hidden layers
- Output layer

The word “**deep**” refers to the depth of the neural network

- More specifically, the number of successive layers used to transform the data



Deep Learning vs. Traditional Machine Learning

Traditional Machine Learning

- Often requires manual feature engineering
 - Human expertise needed to select or design important features
- Can work well with smaller datasets
- Usually requires less computational power
- Often easier to interpret and faster to train

Deep Learning

- Learns useful features automatically from raw or minimally processed data
- Usually benefits from a large amount of training data
- Requires greater computational resources
 - GPUs / high-performance computing
 - Performance can continue to improve as data and computing resources increase

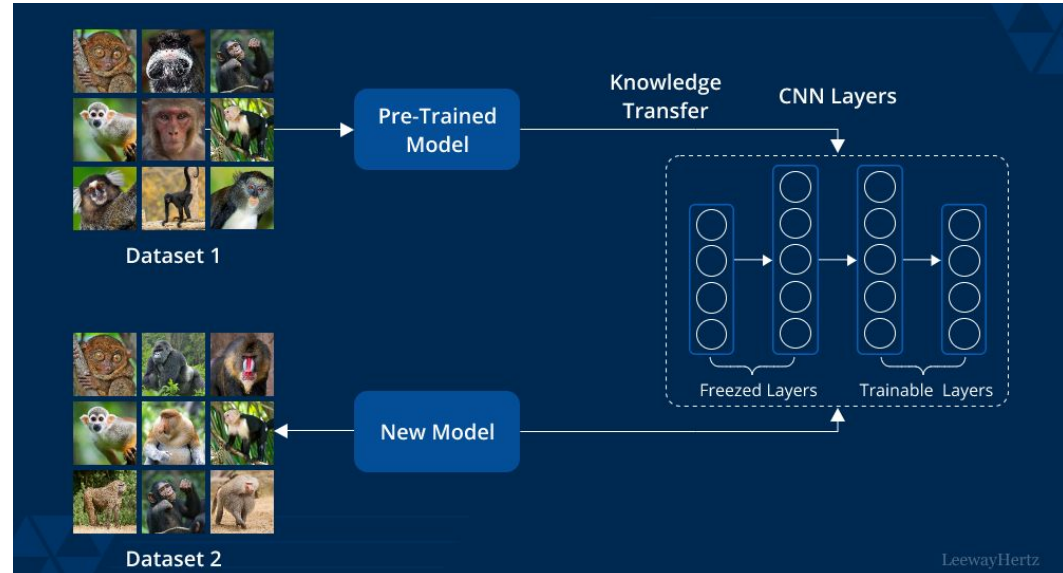
Transfer Learning

Basic idea

- A model trained on one task is adapted (transferred) to work on other related tasks
- **Transferring knowledge** learned from a source to other learning tasks

Examples

- Pretrained models on large image data sets (e.g., VGG, ResNet)
- Pretrained language models such as BERT, GPT



Batch Learning vs. Online Learning

Updating prediction model based on whether they can learn incrementally as new data becomes available

- Critical for big data analytics

Batch/Offline Learning

Observe a **batch** of training data
 $\{(X_j, Y_j) \mid j = 1 \dots N\}$

- Learn a model from them
- Predict new samples accurately

Online/Adaptive Learning

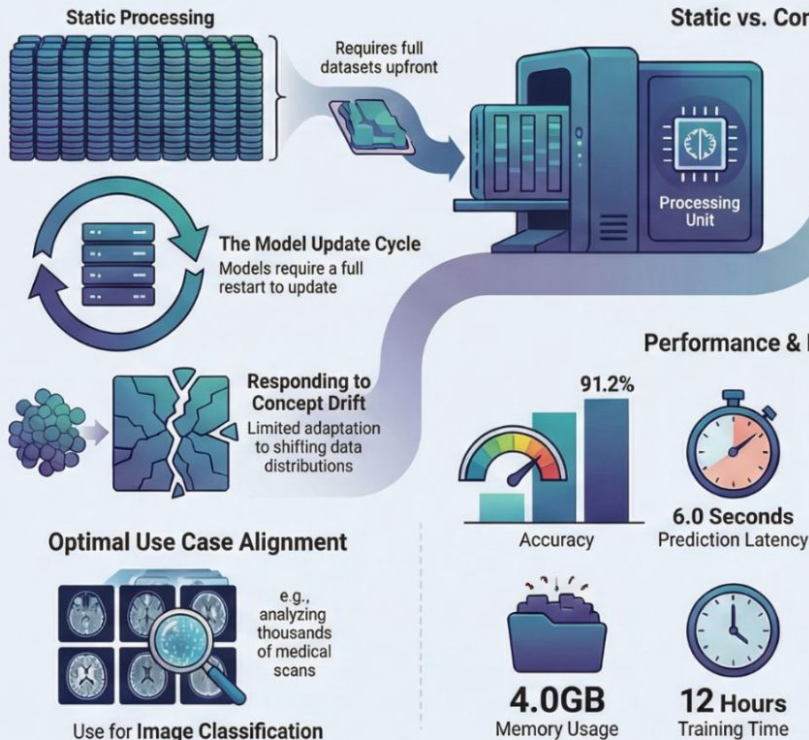
Observe a **sequence** of data
 $(X_1, Y_1), \dots, (X_j, Y_j)$

- Learn a model **incrementally** as instances come
- Make the sequence of online predictions accurately

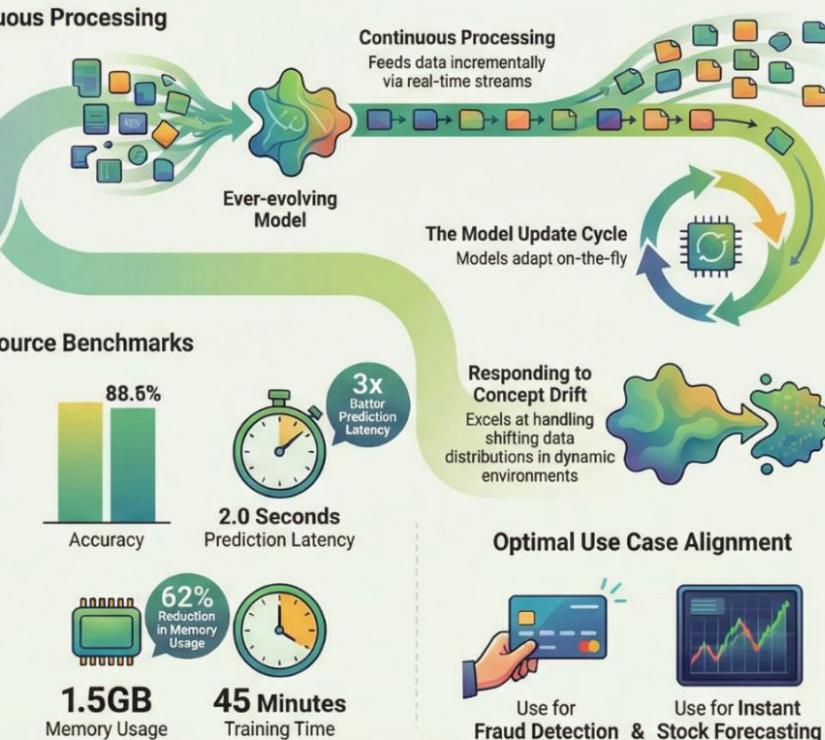
AI Training Paradigms: Batch Learning vs. Online Learning

Machine learning models are trained using two primary paradigms: Batch learning, which processes entire datasets in bulk, and Online learning, which updates incrementally as data arrives. This infographic highlights the trade-offs between precision, resource efficiency, and real-time adaptability.

BATCH LEARNING (Offline)



ONLINE LEARNING (Incremental)



Transfer Learning vs. Batch/Online Learning

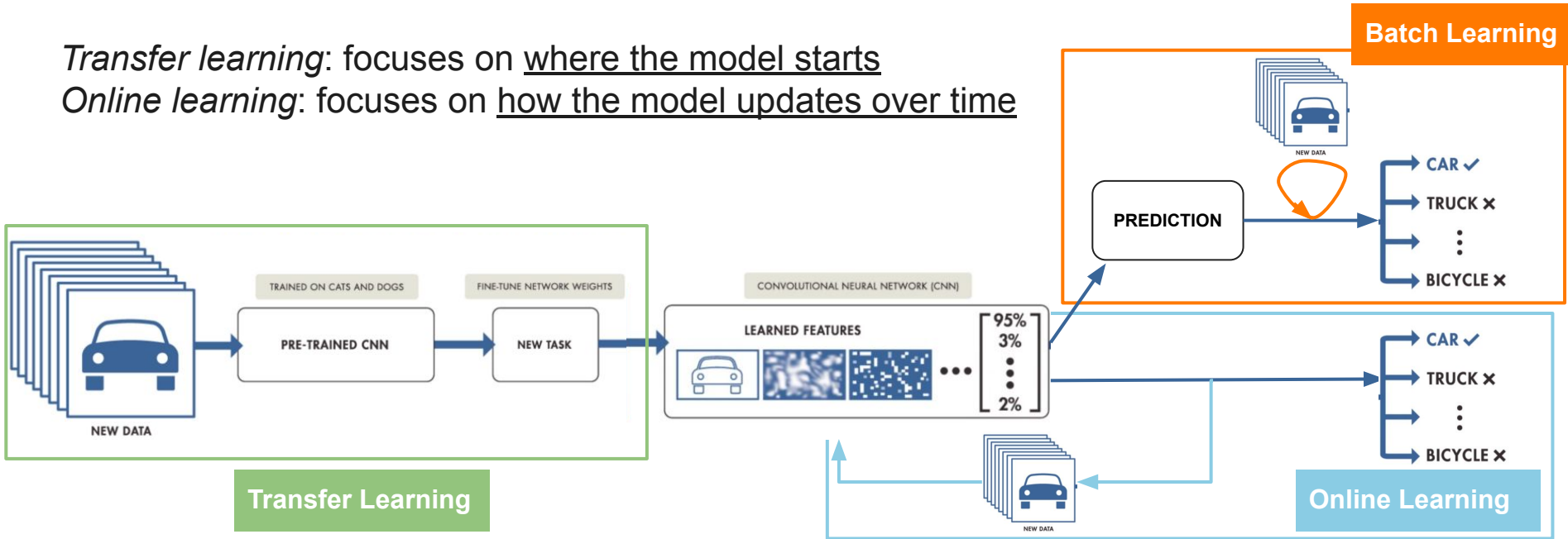
Transfer learning is NOT a type of online learning

- These two are NOT mutually exclusive categories

A transfer-learning model can use either batch learning or online learning

Transfer learning: focuses on where the model starts

Online learning: focuses on how the model updates over time



Ensemble Learning

Basic idea

- Use multiple learning algorithms to obtain better predictive performance

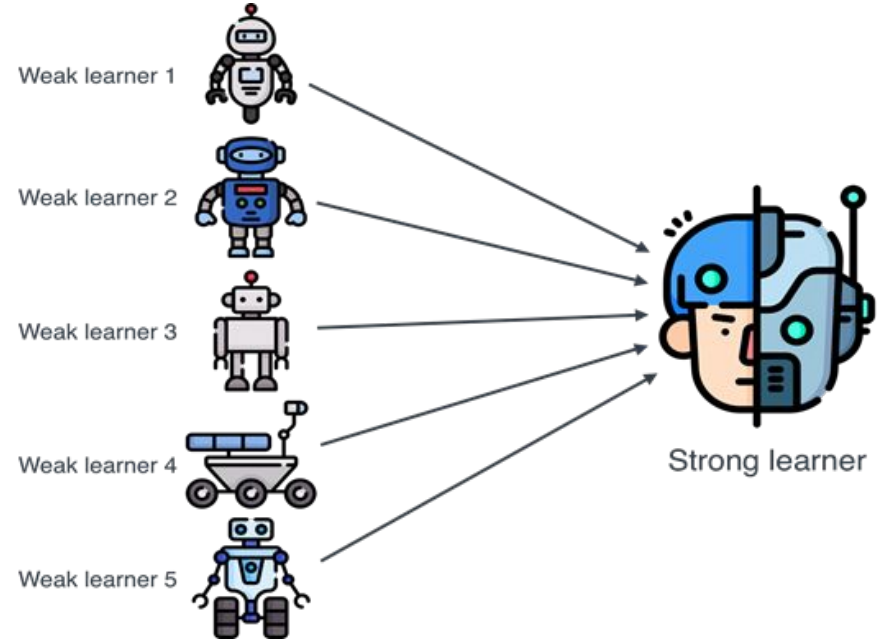
Examples

- Random forests (bootstrapping)
 - Learning from multiple decision trees
- Bayes optimal classifiers

Challenges

- Incompatibility of data and models between different algorithms due to the limitation of knowledge representation
- Computing resources

More thoughts, the better?



Summary of Machine Learning

Supervised Learning

- Learning by training data through training, also called inductive learning (generalization)
- Two types of problems: classification and regression

Unsupervised Learning

- Clustering or finding associations in any data

Semi-supervised Learning

- Mixing supervised and unsupervised learning

Self-supervised learning

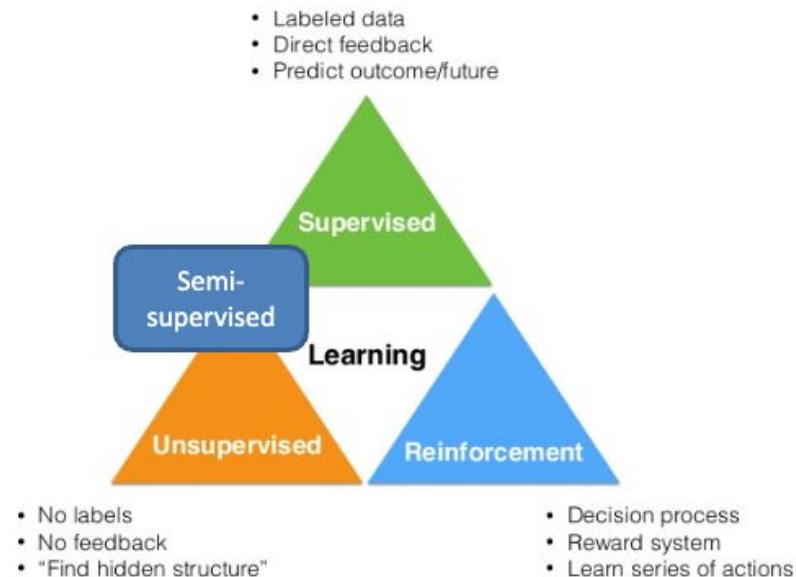
- Extracting some labels from unlabeled data

Reinforcement learning

- Learning rules or tasks by trial-and-error

Other categories

- Deep learning
- Online or adaptive learning
- Transfer learning
- Ensemble learning



Choosing the Right ML Methods

Availability of training data

- *Supervised, unsupervised, or semi-supervised*

Value prediction, classification, clustering, finding associations

- *Regression, classification, clustering, association rule mining*

Finding the **best policy for actions** for optimal decision making

- *Reinforcement learning*

Available computing power

- *Deep learning*

Big data

- *Online (real-time) or batch learning*

"No Free Lunch" :(

D. H. Wolpert. The supervised learning no-free-lunch theorems. In *Soft Computing and Industry*, pages 25–42. Springer, 2002.

Our model is a simplification of reality



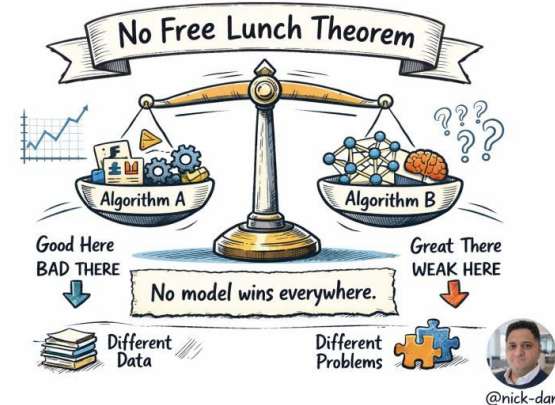
Simplification is based on assumptions (model bias)



Assumptions fail in certain situations

Roughly speaking:

"No one model works best for all possible situations."



ML-related roles in the industry

ROLE		RESPONSIBILITIES
Data Engineer		Makes the appropriate data available for data scientists; focuses on data integration, modelling, optimization, quality and self-service.
Data Scientist		Identifies use cases, determines appropriate datasets and algorithms, experiments and builds AI models.
AI Architect		Is the glue between data scientists, data engineers, developers, operations (DevOps, DatOps, MLOps) and business unit leaders to govern and scale the AI initiatives.
ML Engineer		Deploys AI models through effective scaling and ensuring production readiness, ensures continuous feedback loop.